

Kuruva Pavan Sai Krishna

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SUMMARY

Computer Science Graduate with hands-on experience in **Software Development**, **AI/ML**, and automation workflows. Built real-time **Computer Vision systems** using **YOLOv8**, web applications using **Firebase**, and automation tools using **Python**. Proficient in **Python, SQL, REST APIs**, with strong interest in **Software Engineering** and **AI-driven applications**.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, C, C++

ML / AI: PyTorch, Scikit-learn, BERT, YOLOv8, CNNs, NLP, Transfer Learning, Computer Vision, Model Fine-tuning, Hyperparameter Tuning, Logistic Regression

Data / Analytics: Pandas, NumPy, OpenCV, BeautifulSoup, Regex, Exploratory Data Analysis (EDA), Feature Engineering, Data Cleaning

Web / Backend: HTML, CSS, Firebase, REST APIs, Gradio, Role-Based Access Control (RBAC)

Tools: Git, GitHub, Google Colab, VS Code, Linux

CS Fundamentals: Data Structures and Algorithms, Database Management Systems, Operating Systems, Object-Oriented Programming

EDUCATION

Indian Institute of Information Technology, Design and Manufacturing, Kurnool

2022 – Present

Bachelor of Technology in Computer Science and Engineering

CGPA: 6.67 / 10

EXPERIENCE

Samsung Innovation Campus | *ML / NLP Intern*

Oct 2025 – Mar 2026 | Remote

- Fine-tuned BERT (bert-base-uncased) on **49,000+ clinical text samples** for 4-class mental health classification (Anxiety, Depression, Normal, Suicidal), achieving **85% overall accuracy** with robust F1-scores across all categories.
- Engineered a custom PyTorch data pipeline with tokenization, attention masking, and AdamW optimizer; applied dropout regularization and learning rate scheduling to improve model generalization.
- Deployed an interactive Gradio web interface for real-time NLP inference, exposing per-class confidence scores and probability outputs for live mental health text screening.

GenAI Lakes | *AI Intern*

May 2025 – Aug 2025 | Remote

- Developed a medical imaging CNN for anorectal malformation detection achieving **98% accuracy** by applying transfer learning and hyperparameter tuning on **3,000+ annotated images**.
- Constructed data preprocessing and augmentation pipelines (rotation, flipping, normalization) to improve model generalization across varied medical scan inputs.
- Built a candidate-joining prediction model achieving **85% accuracy** using logistic regression and feature engineering on structured HR datasets.

Trainity | *Data Analytics Intern*

Apr 2024 – Aug 2024 | Remote

- Performed exploratory data analysis (EDA) on structured business datasets using Python, Pandas, and SQL, delivering insights through visual dashboards and reports for stakeholder review.
- Cleaned and transformed datasets for analytics and reporting pipelines; authored business-oriented case studies on real-world sales and operational data.

PROJECTS

Real-Time Pothole Detection — *Python, YOLOv8, OpenCV*

[GitHub] | Jan – Apr 2026

- Trained a custom YOLOv8 object detection model achieving **93% mAP** on annotated road images; built an automated XML-to-YOLO preprocessing pipeline with augmentation and annotation cleanup.
- Designed a severity estimation module using bounding-box geometry and pixel-depth cues to classify potholes into 3 levels, enabling actionable infrastructure reporting.

Resume Web Scraper for ML Pipelines — *Python, BeautifulSoup, Regex*

[GitHub] | Jun 2025

- Extracted and structured data from **500+ resumes** using regex-based pattern matching across multiple formats, generating a clean ML-ready dataset with automated data cleaning and normalization.

Hostel Management System — *HTML, CSS, JavaScript, Firebase*

[GitHub] | Feb – Apr 2025

- Delivered a full-stack web application managing room allocation, fee tracking, and maintenance requests for **200+ students**; secured access with Firebase Authentication and role-based access control for admin and student roles.